

LLC Resonant DC-DC Converter for High Voltage Applications

Nor Azura, S. Iqbal and Soib Taib

School of Electrical and Electronics Engineering

Engineering Campus, Universiti Sains Malaysia

13800, Nibong Tebal, Pulau Pinang, Malaysia

norazura200584@gmail.com, shahidsidu@hotmail.com and soibtaib@eng.usm.my

Abstract—In this paper, an LLC resonant dc-dc converter is proposed for high voltage applications. Full bridge inverter, LLC resonant tanks, high voltage transformer and full bridge rectifier are comprised in the proposed converter circuit. In operating frequency of proposing converter, output voltage regulation is achieved by narrow varying. The energy trapped in leakage inductance of the transformer and to achieve ZVS feature, LLC resonant circuit is utilized to retrieve. It also increases conversion efficiency. A high voltage transformer is used to step up AC voltage which is then refined by the rectifier. With the equivalent circuit derived by first harmonic approximation (FHA method, input-output response and DC characteristics in frequency domain are obtained. The theoretical equations of the circuit operations are studied in detail to infer. Finally, a design example is implemented out to establish the feasibility of the proposed LLC resonant converter. The simulation results on the essential switching and circuit operation are presented. The robustness of this study is verified by the simulation results of the proposed converter with 300V input and 3.5kV/0.35A output.

Keywords—LLC dc-dc converter, high-voltage, high-frequency

I. Introduction

DC-DC converters are more widely applied in the application such as power systems for computer, battery chargers, telephone equipment [1, 2], electronic ballasts, induction heating [3], high density DC-DC converters, inductive power transfer, plasma display panels (PDPs), photovoltaic solar systems, fuel cells [4], ultracapacitors, electric vehicle [5] and so on. A natural topology choice represents for the high frequency and high voltage applications in the resonant converters [6]. Various advantages of resonant converters over the conventional PWM converter are offered. Some of these advantages include: (1) At higher switching frequencies has a lower switching losses [7] and switching stresses [8], (2) In different application areas, the capability has been applied for operating in different power conversion modes [9], (3) low EMI, and (4) high power density [10]. There are various topologies of resonant converter such as series resonant converter (SRC), parallel resonant converter (PRC) and so on. At low output voltage and high input applications for high efficiency performance [11], and also for high power supplies and high voltage [12], a series resonant converter (SRC) is favored. SRCs are the simplest configuration and well documented [13]. From full load to part load, the SRCs have a

high efficiency. But, the output voltage at the light load is the major disadvantages of SRCs are the difficulty to controlling [14-16]. In the semiconductor devices can be eliminated the turn-on and turn-off losses when PRC has worked with either zero-current turn-off or zero-voltage turn-on. This converter is specially fitted for high power and high frequency operation because due to the reduced switching losses [17]. At switching frequency equal to the resonant frequency, PRC behaves as a constant current source [18]. At switching frequency lower than the resonant frequency, PRC is near to a constant voltage source of the output voltage and has wide load variation, thus it is preferred for voltage regulator applications [19]. PRC possesses high converter gain which requires high turn ratios of a transformer for voltage step-down. This may show a problem in the high-frequency transformer design because at secondary side capacitor has high value [20]. The PRC may have the advantage of power density and uses fewer passive components compared with other resonant converters such as an LLC [21].

Series-parallel resonant converter (SPRC) is combined both series and parallel resonant converters that mostly retained their advantages, while eliminated their disadvantages, also can maintain excellent efficiency when operate over a large load range (no load to full load) and a large input voltage range [22]. When increase the switching frequency, SPRC also provides avoided reducing the efficiency and soft-switching. Thus, for high voltage capacitor charging applications is preferred to use the LLC resonant converter [23]. The advantages SPRC are: (1) The converter is like an SRC under full load and like PRC under light load is allowed for voltage conversion characteristics, and (2) an inherent short circuit protection [24]. ZVS at the primary side switches and ZCS at the secondary side rectifiers can achieved in the LLC resonant converter [25], and possesses high efficiency at high input voltage. It also has some of unique and improvements characteristics over other topologies [26]. LLC resonant converter is able to operate both step-up and step-down modes [27] and function well with additional wide input voltage range by using frequency control [28]. Compare with conventional resonant converters, an LLC resonant converter has many advantages such as: (1) A narrow switching frequency range, making wide output regulation range, (2) At the switching loss with low turn-off

current and ZVS capability in the entire load range is very low, (3) to achieve ZVS the use of all extremely important elements, and (4) zero current switching (ZCS) on secondary rectifier and low voltage stress. On rectifier diodes limited to two times output voltage, low voltage stress with no regulation range, (2) At the switching loss with low turn-off current and ZVS capability in the entire load range is very low, (3) to achieve ZVS the use of all extremely important elements, and (4) zero current switching (ZCS) on secondary rectifier and low voltage stress. On rectifier diodes limited to two times output voltage, low voltage stress with no secondary inductor [29, 30]. In high current, low voltage application, LLC resonant converter is widely done and also for low current, high voltage application [31].

In Section II, circuit descriptions of the proposed LLC resonant converter are explained detail. The analysis of converter for DC characteristics is presented in Section III. In Section IV, design examples of proposed converter are introduced. Then, the simulation results based on LLC resonant converter for high-voltage application are described in Section V and in Section VI, the conclusion are explained.

II. Circuit Description

The three stages as depicted in Figure 1 are square wave generator; resonant tank and rectifier circuit consist in LLC resonant topology.

- At switches S_1 , S_4 and S_2 , S_3 alternately and usually MOSFETs to be capable to work at high frequencies in the square wave generator creates a square wave voltage driving with 50% duty cycle for each switch. Usually introduced a small dead time between the consecutive transitions. A half-bridge and full-bridge type can be used to make the square wave generator stage.
- The leakage inductance, L_r , the magnetizing inductance, L_m of the transformer and a capacitor, C_r are consists in the resonant tank. The higher harmonic current in the resonant tank can be filtered. Basically, in the resonant tank can allow only sinusoidal current to flow through, even though used a square wave voltage. MOSFETs can allow to be turned on with zero voltage when the voltage used to the resonant tank is current lagging, I_L . MOSFET is zero, while the voltage across through the anti-parallel diode with current flowing as shown in Figure 3 will turn on the MOSFETs.
- A DC voltage is generated in the rectifier circuit when the diode is applied to rectify AC current and the capacitor to remove a ripple. A center-tapped or full-wave bridge configuration can be used in the rectifier circuit with the capacitive output filter. However, for high voltage applications is more suitable to use full-wave bridge configuration. It is because full-wave rectifier has more advantages compare to half-wave rectifier such as: (1) has any specified DC component or purely produces DC for an output voltage or current, (2) compare to the half-wave rectifier, the average (DC) output voltage is higher and (3) also producing a

smoother output waveform because the full-wave rectifier has much less ripple.

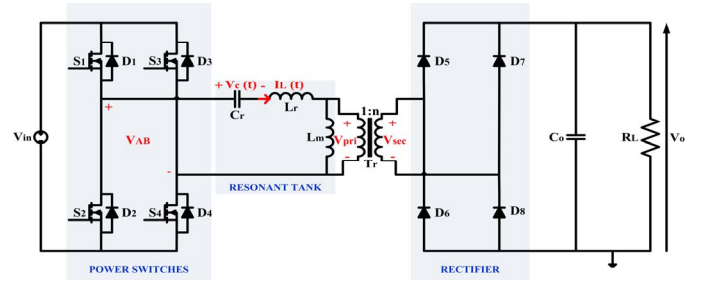


Figure 1: Schematic diagram of full-bridge LLC resonant circuit.

The operating waveforms at below resonance are illustrated in Figure 2. In driving pulse width before to the end, resonant current has decreased to the value of magnetizing current, making magnetizing current continues even through the power transfer has stopped. Primary ZVS can still achieve and the soft commutation can be obtained of the rectifier diodes on the secondary side when operation below the resonant frequency. The discontinuous current mode at the secondary-side diodes and more circulating current with the same quantity of energy has delivered to the load is required in the resonant circuit. In higher conduction losses has an additional current result in both the primary and secondary sides. However, if the switching frequency becomes too low, should be noted one characteristic at the primary ZVS may be misplaced.

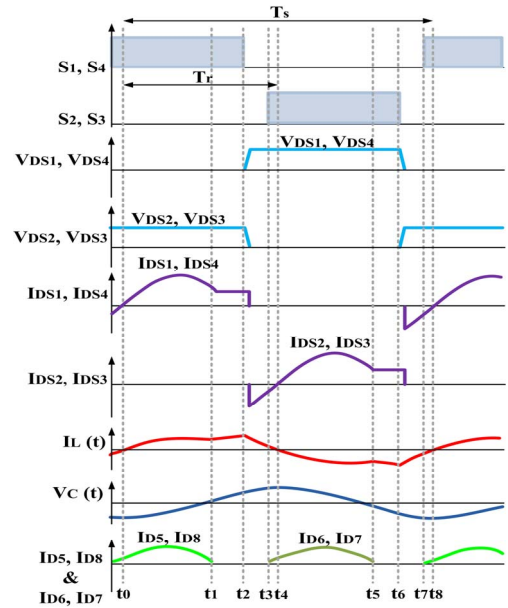


Figure 2: Key operating waveform of the converter at below resonance.

III. Analysis of Converter

DC characteristics of the LLC resonant converter are discussed in this paper. The equivalent circuit of the LLC topology is derived by applying first harmonic approximation

(FHA) acquires an input-output response in the frequency domain. A full-bridge converter as shown in Figure 1 is to analyze at the LLC resonant circuit. Its equivalent circuit can be represented in

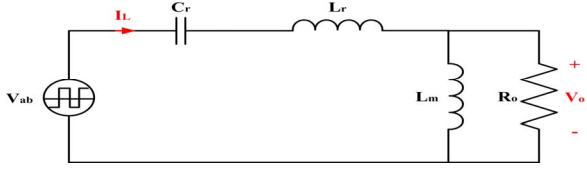


Figure 3: Equivalent circuit of LLC resonant circuit.

Figure 3, where $R_{o(ac)}$ are applied to the LLC of the transformer, T_r and V_{ab} at primary side in the equivalent load resistance. The equivalent load resistance, $R_{o(ac)}$ can be realized to equivalent circuit of LLC resonant circuit,

$$R_{o(ac)} = \frac{8R_L}{(n\pi)^2} \quad (1)$$

where n is the turns ratio of transformer, T_r and R_L is load resistance. To analyze the resonant frequency, f_{r1} of C_r and L_r in series of the LLC resonant topology is derived from

$$f_{r1} = \frac{1}{2\pi\sqrt{L_r C_r}} \quad (2)$$

Likewise, as follows the resonant frequency, f_{r2} of C_r and $(L_r + L_m)$ in series is shown:

$$f_{r2} = \frac{1}{2\pi\sqrt{(L_r + L_m) C_r}} \quad (3)$$

In addition, the LLC resonant circuit is used, can be approximated by maximum value, $V_{ab(max)}$ and the fundamental component, V_{ab} according to the square wave and the Fourier theorem can be derived from

$$V_{ab(max)} = \frac{4}{\pi} V_{in} \quad (4)$$

where V_{in} is input voltage of full-bridge converter. Thus, rms value of V_{ab} can be defined as

$$V_{ab(rms)} = \frac{2\sqrt{2}}{\pi} V_{in} \quad (5)$$

As depicted in Figure 1, to analyze the LLC resonant circuit of full-bridge converter of input to the output transfer, G is derived from

$$G(s) = \left| \frac{V_o}{nV_{in}} \right| \frac{(sL_m // R_{o(ac)})}{sL_r + \frac{1}{sC_r} + (sL_m // R_{o(ac)})} \quad (6)$$

From (6) when the proposed converter is controlled at different loads, it can be determined that the transfer ratio, G will be varied. To achieve a constant transfer ratio, G , the proposed one must determine the switching frequency for generating constant voltage output. Therefore, transfer ratio, G can be rewritten as

$$G(\omega) = \left| \frac{V_o}{nV_{in}} \right| \frac{1}{\sqrt{\frac{1}{\frac{L_r}{L_m} \frac{L_r C_r}{\omega_s^2} \left(\frac{\omega_s}{\omega_{r2}} \right)^2} + \left[\frac{\pi^2}{8} Q \left(\frac{\omega_s}{\omega_{r2}} - \frac{\omega_{r1}}{\omega_s} \right)^2 \right]}} \quad (7)$$

where $\omega_s = (2\pi f_s)$ is the angular frequency of switching frequency, f_s , Q is quality factor, $\omega_{r1} = (2\pi f_{r1})$ is that of frequency f_{r1} , and $\omega_{r2} = (2\pi f_{r2})$ is that of frequency f_{r2} . Quality factor, Q in (7) can be shown by

$$Q = \frac{n^2 \sqrt{\frac{L_r}{C_r}}}{R_L} \quad (8)$$

When ratio $L_n = (L_m/L_r)$ and frequency gain $f_n = (f_s/f_{r1})$ are described, transfer ratio G can be rewritten as

$$G = \left| \frac{V_o}{nV_{in}} \right| \frac{1}{\sqrt{\left(\frac{1}{L_n} + 1 - \frac{1}{L_n f_n^2} \right)^2 + \frac{\pi^4}{64} Q^2 \left(\frac{1}{f_n} - f_n \right)^2}} \quad (9)$$

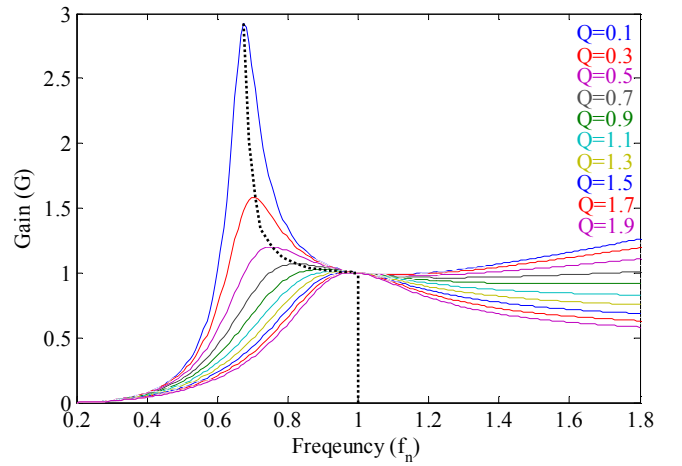


Figure 4: Plots of DC characteristic of the LLC resonant converter.

DC characteristics according to (9) in the LLC resonant converter can illustrated in Figure 4 respectively and winding turn ratio, $n=1$ at the transformer. At the voltage step-down operation, $G < 1$, frequency variation is needed very wide range where switching frequency, f_s is higher than frequency resonant, f_{r1} , in the frequency area. The complicated and impractical design task at the output voltage and power control area and yields, as a result, in the restriction for the winding turn ratio of the high frequency transformer. At the voltage step-up operation, $G \geq 1$, the curve, G has more correlation with the LLC resonant converter when switching frequency, f_s almost approaching resonant frequency, f_{r1} . A DC characteristic according to equation 9 as shown in Figure 5 is utilized in the proposed LLC resonant converter.

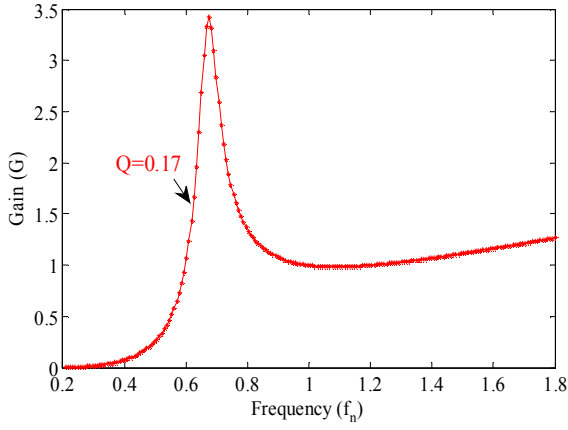


Figure 5: DC characteristic for proposed LLC resonant converter.

IV. Design Example

The design of the example converter with specification is given as follows. The design of the high voltage and low current output is focused.

A. Design Specification

The specifications have been established for the simulation of a proposed LLC converter as follows:

- Input voltage: $V_s = 300\text{-}400\text{V}$
- Output voltage: $V_o = 4\text{kV}$
- Output current: $I_o = 0.5\text{A}$
- Resonant frequency, $f_s = 50\text{KHz}$

B. Selecting the Switching Frequency, (f_s)

For certain applications are usually defined, so that the switching frequency acceptable. As an example, a switching frequency 150 kHz at most of-line AC/DC applications is required for normal operation. For unique applications, a different frequency range is demanded and also various factors that may require consideration. As is known, ZVS efficiency and the less important switching losses become and bulkier the converter because the lower switching frequency. Forming the LLC converter is less attractive and the conduction losses become dominant.

In comparison with hard-switched converter, especially for higher switching frequency makes the advantages of the LLC converter more pronounced. Additional factors may have to be reviewed if the switching frequency is very high, such as an additional switching loss despite MOSFET ZVS and component availability, as example magnetic-core losses. The series resonant frequency, f_{r1} is above the switching frequency as shown in Figure 2 at the below resonance operation. LLC resonant converter is required to be around 50 kHz for the switching frequency, so that the series resonant frequency, f_{r1} have to be a larger than 50 kHz should be selected. The series resonant frequency f_{r1} is 61.25 kHz are selected in this proposed LLC resonant converter.

C. Selecting the Transformer Turns Ratio, (n)

To consider isolation requirement and high step-up ratios for the proposed converter to generate output voltage, a full-

bridge converter and high frequency transformer is set. Since the LLC resonant converter is needed to function at the below resonance operation, the condition should meet the expectations as follows

$$n = \frac{N_p}{N_s} = \frac{V_{in-max}}{V_o} \quad (10)$$

where V_{in-max} is the maximum input voltage.

From equation (10), the conditions are followed for the transformer turn ratio is derived

$$n = \frac{V_{in-max}}{V_o} = 0.1$$

Therefore, the transformer turn ratio is selected are 30:300.

D. Selecting of Resonant Tank (L_m , L_r and C_r)

As shown in (11), the condition of the magnetizing inductor, L_m must meet and then the primary switches can turn ON under ZVS.

$$L_m \leq \frac{nV_o T_s t_{d,min}}{8C_{oss} V_{in}} \quad (11)$$

The minimum dead time $t_{d,min}$ is 1.5μs and the equivalent output of power MOSFETs C_{oss} is 90nF such as in the illustrative example. Hence, can be expressed of the maximum inductance, $L_{m,max}$ of as follows:

$$L_{m,max} = \frac{nV_o T_s t_{d,min}}{8C_{oss} V_{in}} = 56\mu\text{H}$$

By using the definition of inductor ratio, also can determine the maximum inductance, $L_{r,max}$ of as follows:

$$L_{r,max} = \frac{1}{L_n} L_{m,max} = 45\mu\text{H} \quad (12)$$

From (2), the resonant capacitor, C_r can be expressed as

$$C_r = \frac{\left(\frac{1}{2\pi f_{r1}}\right)^2}{L_r} = 150\text{nF}$$

V. Simulation Results

Table 1 presents a summary of the LLC resonant converter of the electrical specifications and component parameters based on the previous design. Operational modes have been demonstrated in OrCAD PSpice simulation software for the LLC resonant converter.

Table 1: Electrical specifications and component parameters of the design example

Electrical Specifications	
Input Voltage, V_{in}	300 V
Output Voltage, V_o	3.5 kV
Output Current, I_o	0.35 A
Resonant Frequency, f_{r1}	61.25 kHz
Secondary Resonant Frequency, f_{r2}	40 KHz
Switching Frequency, f_s	50 kHz

Component Parameters	
Resonant Inductor, L_r	45 μH
Magnetizing Inductor, L_m	56 μH
Resonant Capacitor, C_r	150 nF
Transformer Turn Ratio, $1:n$	30:300

Figure 6 shows the simulation waveforms at the resonant tank for the voltage applied, V_{ab} and the current flowing, I_L which through it. The operation condition is: input voltage 300V, output 3.5kV/0.35A and switching frequency 50 kHz. The theoretical waveform as shown in Figure 2 corresponds well with the simulation waveform as shown in Figure 6 can be seen. Figure 7 shows the measured voltage gate-source, V_{GS} and voltage drain-source, V_{DS} of switches S_1 and S_4 at the turn on transient. ZVS can achieve in the proposed LLC resonant converter from Figure 7.

Figure 8 shows the capacitor voltage, V_C and inductor current, I_L of the resonant tank. From Figure 8, when the input voltage decreases, so the stresses increase at the resonant tank can be seen. Figure 9 shows that ZCS together with the DCM current, I_S are commutated at the secondary-side diode. It can be seen that the ZCS characteristics can be achieved at the output diodes. Therefore, that proposed LLC resonant converter actually confirmed operate under soft switching conditions. Figure 10 shows the output voltage, V_o and current, I_o , that the output ripple is very low within 1% as illustrated.

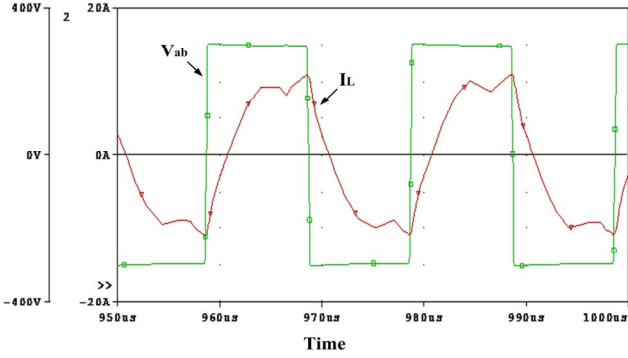


Figure 6: The voltage, V_{ab} and current, I_L waveform at the resonant tank.

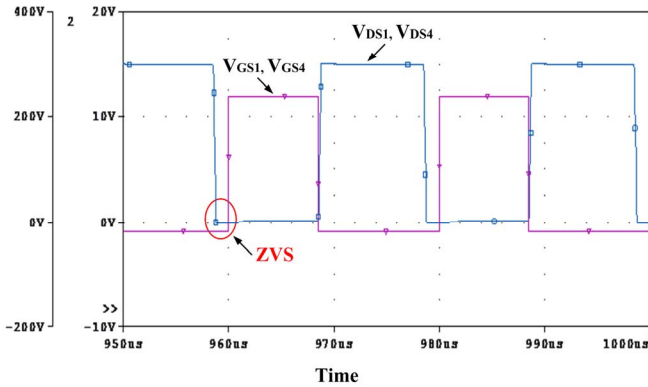


Figure 7: The gate-source, V_{GS} and drain-source, V_{DS} voltages waveform at power switches, S_1 and S_4 .

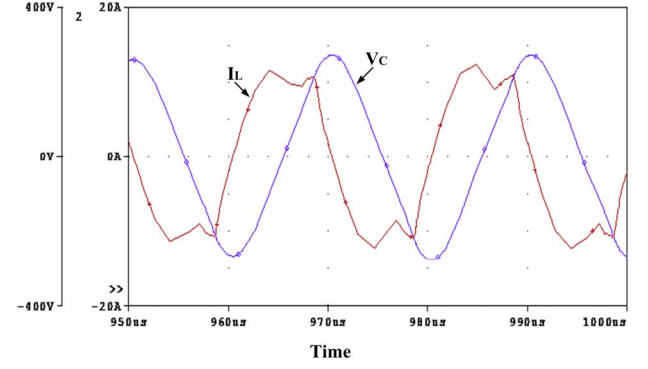


Figure 8: The capacitor voltage, V_C and inductor current, I_L waveform a flowing through resonant tank.

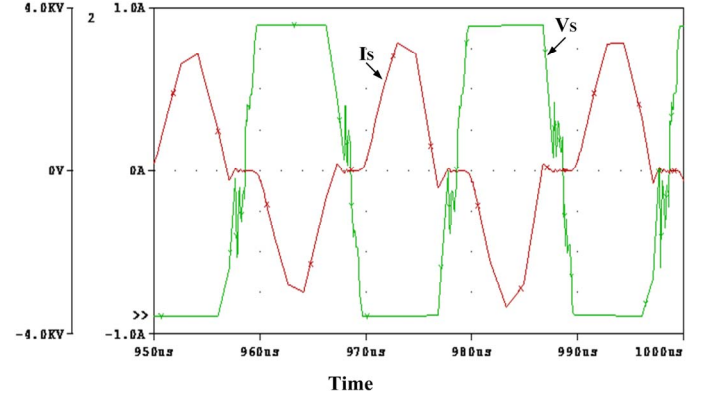


Figure 9: The voltage, V_S and current, I_S waveform at secondary-side of transformer.

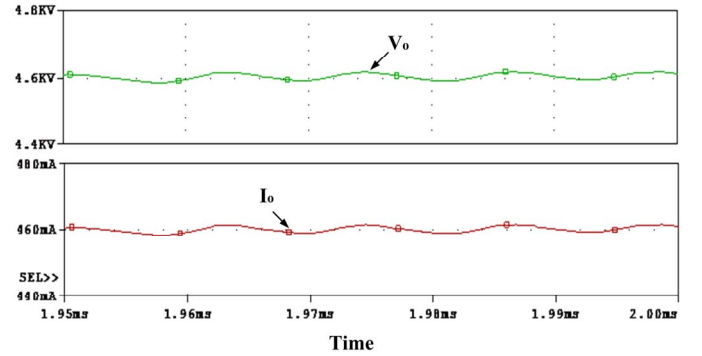


Figure 10: The measured output voltage, V_o and output current, I_o at maximum output.

VI. Conclusion

This paper has proposed full-bridge LLC resonant converter suitable for high voltage application. The power switches are operated under the ZVS condition and the transformer leakage inductance at the LLC resonant can be utilized. The LLC resonant converter uses high frequency transformer to achieve high step-up voltage ratio with high turns ratio. The input voltage of 300V boosts up to 3.5kV in the LLC resonant converter by fixed duty ratio operating. The proposed converter is analyzed in detail and also the circuit parameters

and design equations has been presented and obtained. Therefore, operational principle and steady-state analysis has been proposed. This converter has high-efficiency because the LLC resonant converter operates in soft switching mode with resonance. The proposed LLC converter can also be applied to other applications.

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